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Respiratory humidification system

Abstract

A system provides warm, humidified gas to a patient via a patient interface. Horizontal connections can be used between the humidification chamber and conduit. To reduce the likelihood of condensate flowing back to the humidification chamber, or dead space or gases recirculation regions occurring within the gases flow path, a raised portion is positioned inside of the flow path to improve flow characteristics and to provide a barrier for condensate back flow. The raised portion also reduces the amount of condensate that is formed in the system and provides better flow characteristics for sensing purposes.

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

(1) Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57.

BACKGROUND

Technical Field

(2) The present disclosure generally relates to humidification systems for providing humidified respiratory gases to patients. More particularly, the present disclosure relates to features that improve the performance of humidification systems.

Description of the Related Art

(3) Respiratory humidification systems deliver heated and humidified gases to a patient via a patient interface. Patient interfaces can include a face mask, oral mask, nasal mask, nasal cannula, a combination of oral and nasal mask, tracheal mask, or the like. Gases leaving the humidification chamber to be delivered to a patient are at a temperature and relative humidity that may mimic the transformation of air that occurs as it passes through the nose to the lungs. This promotes efficient gas exchange and ventilation in the lungs, which aids defence mechanisms in the airways and increases patient comfort during treatment. Condensation within the humidification system can occur when gases with high humidity come into contact with a component having cooler walls.

SUMMARY

(4) Delivery of humidified gases to patients exists in the prior art, however an aspect of at least one of the embodiments disclosed herein includes the realisation that there are problems with the delivery of this humidified gas. Horizontal connectors that connect between a humidification chamber and a conduit may allow condensate from the conduit or patient interface to run back into the humidification chamber in use. Elbow ports can be used to form a horizontal connection between a humidification chamber and a conduit, which can cause recirculation to occur resulting in stagnant zones, which cause greater heat loss to occur in these regions. This results in regions of low temperatures, which are more likely to form condensate. Condensate may also form at the connector, as this is an unheated section of the gas pathway. Condensate at the connector may form at the inner surface of the outer wall, before pooling at the base of the conduit, where it may flow back to the humidification chamber.

(5) A system is disclosed which may improve the fluid characteristics of the respiratory gas. Thus, it may reduce the amount of condensate. In some configurations, certain aspects can reduce the amount of condensate that flows back to the humidification chamber relative to the prior art. In some configurations, the angle of the elbow port can be increased to greater than 90° to reduce the recirculation that causes stagnant zones to occur and thereby reduce the condensate that forms in the system. Some embodiments include an insert or other structure shaped as a raised portion, which can be located near the sharp corner of the elbow port, to fill at least some of the dead space that is formed due to the recirculation of gases that occurs. The insert may reduce the likelihood of recirculation occurring in this region and thus may reduce the likelihood of stagnant zones forming. Therefore, regions of low temperature and low velocity gas are reduced, which reduces the amount of condensate that is generated. A raised portion may also act as barrier to any condensate that is formed and, as such, reduce the likelihood of condensate flowing back to the humidification chamber. A raised portion may be added to the port of the humidification chamber, may be part of the connector that inserts when the connector is connected to the port, or may be part of the conduit.

(6) Some embodiments may comprise either the humidification chamber outlet port or the conduit being shaped in such a way that, or including geometries such that, gravity may reduce the likelihood of condensate reaching the humidification chamber.

(7) Therefore, the improved system may reduce condensate formation between the outlet port of

the humidification chamber and the proximal conduit end by improving flow characteristics. The improved system may provide mechanisms to reduce the likelihood of condensate flowing back to the humidification chamber.

(8) In some configurations, a respiratory humidification system comprises a humidification chamber that comprises an outlet port; a conduit that comprises a connector configured to engage with the outlet port such that the conduit forms a gases flow path from the humidification chamber; and a raised portion located in the gases flow path directly adjacent to a sharp corner disposed along the gases flow path within a region defined between the body of the humidification chamber and the conduit.

(9) In some such configurations, the raised portion inhibits liquid flowing from the conduit to the humidification chamber in use.

(10) In some such configurations, the raised portion fills or at least partially fills a dead space region in the gases flow path in use.

(11) In some such configurations, the outlet port comprises a vertical section and a horizontal section and the connector is configured to engage with the horizontal section.

(12) In some such configurations, the raised portion is located within the horizontal section of the outlet port.

(13) In some such configurations, the outlet port comprises a vertical section, the connector comprises a vertical section and a horizontal section, and the vertical section of the connector is configured to engage with the outlet port.

(14) In some such configurations, the raised portion is located within the horizontal section of the connector.

(15) In some such configurations, the raised portion is directly attached to the outlet port.

(16) In some such configurations, the raised portion is directly attached to the connector.

(17) In some such configurations, the raised portion is attached to the conduit.

(18) In some such configurations, the conduit comprises one or more heating elements.

(19) In some configurations, a respiratory humidification system comprises a humidification chamber configured to contain a volume of liquid. The humidification chamber has an outlet port. A conduit is connectable to the outlet port of the humidification chamber. The conduit and the outlet port of the humidification chamber are connected by a connector. A gases flow path is defined from an entrance to the outlet port of the humidification chamber to an outlet of the conduit. A sharp corner is positioned along the gases flow path at a location where the gases flow path makes an abrupt change in direction. A raised portion is located in the gases flow path directly adjacent to the sharp corner.

(20) In some such configurations, the raised portion is located immediately downstream from the sharp corner.

(21) In some such configurations, the raised portion is located within a region of the gases flow path that would be a recirculation region without the raised portion present.

(22) In some such configurations, the raised portion forms a portion of the outlet port.

(23) In some such configurations, the raised portion forms a portion of the connector.

(24) In some such configurations, the raised portion forms a portion of the conduit.

(25) In some such configurations, the raised portion fills at least a lower portion of at least one of a horizontal portion of the outlet port, a conduit or a connector such that condensate is less likely to flow from the conduit into the humidification chamber.

(26) In some such configurations, the raised portion has a tapered edge.

(27) In some such configurations, the raised portion has a straight edge.

(28) In some such configurations, the raised portion is asymmetrical along its length.

(29) In some such configurations, the raised portion is symmetrical along its length.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and other features, aspects, and advantages of the present disclosure will be described with respect to the following figures, which are intended to illustrate and not to limit the preferred embodiments.
- (2) FIG. 1 is a schematic view of an example embodiment of a patient gas supply system.
- (3) FIG. 2 is a perspective view of a humidifier useable in the system of FIG. 1.
- (4) FIG. 3 is an enlarged section view of a portion of the humidifier of FIG. 2 and showing a first configuration of a raised portion within the gases flow path.
- (5) FIG. 4 is a view of a portion of a humidifier chamber useable with the humidifier of FIG. 2 and showing a second configuration of a raised portion within the gases flow path.
- (6) FIGS. 5 and 6 show cross-sections of raised portions.
- (7) FIG. 7 shows a velocity profile of a humidification chamber with an elbow port for a 5 L/min flow.
- (8) FIG. 8 shows a temperature profile of a humidification chamber with an elbow port for a 5 L/min flow.
- (9) FIG. 9 shows a velocity profile of a humidification chamber with an elbow port for a 30 L/min flow.
- (10) FIG. 10 shows a temperature profile of a humidification chamber with an elbow port for a 30 L/min flow.
- (11) FIG. 11 shows a velocity profile of a humidification chamber with an elbow port for a 60 L/min flow.
- (12) FIG. 12 shows a temperature profile of a humidification chamber with an elbow port for a 60 L/min flow.
- (13) FIGS. 13(a)-13(d) show different configurations of gas flow paths with three different raised portions illustrated in FIGS. 13(b)-13(d).
- (14) FIGS. 14(a)-14(d) illustrate flow velocity contours through the configurations of FIGS. 13(a)-13(d) for a 5 L/min flow.
- (15) FIGS. 15(a)-15(d) illustrate reverse flow vectors overlaid onto the flow velocity contours of FIGS. 14(a)-14(d).
- (16) FIGS. 16(a)-16(c) illustrate show streamlines coloured by velocity magnitude and indicating paths taken by a zero mass particle if that particle were traveling with the flow.
- (17) FIG. 17 is a plot illustrating the flow velocity components in a 5 LPM flow relative to the direction of flow at the same cross section for each of the configurations of FIGS. 13(a)-13(d).
- (18) FIG. 18 is a plot comparing pressure magnitudes taken along a horizontal line that extends along an axis of the configurations of FIGS. 13(a)-13(d).
- (19) FIGS. 19(a)-19(d) show additional humidification chamber embodiments designed to impede condensate flowing back into the humidification chamber.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

- (20) With reference to FIG. 1, a system **100** is illustrated that can be used to supply heated and/or humidified gases flow to a patient or other user. The system **100** can be configured to be a continuous, variable, or bi-level positive airway pressure (PAP) system, invasive or non-invasive respiratory assistance system, high flow respiratory therapy system, a surgical insufflation system, or a system providing other forms of medical gases. The system **100** may be readily adapted for other applications involving the supply of a heated and/or humidified gas flow to a user or patient, including but not limited to laparoscopy, ventilation support, and the like. Such applications may use alternative gases, operating parameters (e.g., flow, pressure, temperature, or humidity) and patient interfaces.

- (21) In the system **100**, dry or relatively dry gases pass from a gases source **102** to a humidifier **104**. The gases source **102** may be, for example, a ventilator or a blower.
- (22) The humidifier **104** conditions the dry or relatively dry gases. For example, the humidifier **104** can supply heat or humidity to the dry or relatively dry gases.
- (23) An inspiratory tube **106** is used to deliver the conditioned gases to a patient P. In the illustrated configuration, a distal end **110** of the inspiratory tube **106** connects to a port **112** of the humidifier **104**. Thus, the humidifier **104** supplies conditioned gases to the inspiratory tube **106**.
- (24) In the illustrated configuration, the conditioned gases flow through the inspiratory tube **106** to a Y-piece **114**. A patient interface **116** (e.g., a mask) receives the conditioned gases from the Y-piece **114** and the patient interface **116** supplies the conditioned gases to the patient P. Any suitable patient interface may be incorporated. Patient interface is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art (that is, it is not to be limited to a special or customized meaning) and includes, without limitation, masks (such as tracheal mask, face masks, oral masks, nasal masks, a combination of oral and nasal mask, for example), cannulas (nasal cannulas, for example), and nasal pillows.
- (25) In the illustrated configuration, an expiratory tube **120** also connects to the patient interface **116** through the Y-piece **114**. The expiratory tube **120** may be configured to move exhaled gases away from the patient P. In the illustrated configuration, the expiratory tube **120** returns exhaled gases from the patient P to the gases source **102**.
- (26) In illustrated configuration, dry or relatively dry gases enter the gases source **102** through a vent **122** or other inlet. A fan, blower or other flow generator **124** may improve gas flow into the gases source **102** by drawing air or other gases through the vent **122**. The flow generator **124** may be, for instance, a variable speed fan.
- (27) An electronic controller **126** controls the speed of the flow generator **124**. In particular, the function of the electronic controller **126** may be controlled by an electronic master controller **130** in response to inputs to the master controller **130** and a user-set predetermined required value (e.g., a preset value) of pressure or fan speed. The value can be set using an input component, such as a dial **132**, for example but without limitation.
- (28) The humidifier **104** comprises a humidification chamber **134**. The humidification chamber **134** comprises the port **112**. The body of the humidification chamber **134** can contain a volume of water W or other suitable humidifying liquid. The humidification chamber **134** is removable from the humidifier **104** after use to allow the humidification chamber **134** to be more readily sterilized or disposed.
- (29) The body of the humidification chamber **134** may be formed from a non-conductive glass or plastics material. The humidification chamber **134** may comprise conductive components. For instance, the humidification chamber **134** may comprise a highly heat-conductive base (for example, an aluminum base). The heat-conductive base can contact or associate with a heater plate **136** on the humidifier **104**.
- (30) The humidifier **104** may also include electronic controls. In the illustrated configuration, the humidifier **104** includes the master controller **130**. The master controller **130** can be an electronic, analog, or digital master controller. The master controller **130** may be a microprocessor-based controller executing computer software commands stored in associated memory. In response to humidity or temperature values provided via a user interface **140**, for example, and other inputs, the master controller **130** determines when (or to what level) to energize the heater plate **136** to heat the water W within the humidification chamber **134**.
- (31) A sensor probe **142** may connect to the inspiratory tube **106** near the Y-piece **114**, or directly to the Y-piece **114** or the patient interface **116**. The sensor probe **142** monitors the temperature of the flow of gases near or at the patient interface **116**. A heating filament (not shown) may be used to adjust the temperature of the patient interface **116**, the Y-piece **114**, and/or the inspiratory tube **106** to raise or maintain the temperature of the flow of gases above the saturation temperature, thereby

reducing the opportunity for unwanted condensation.

(32) In FIG. 1, exhaled gases are returned from the patient interface **116** to the gases source **102** via the expiratory tube **120**. The expiratory tube **120** may have a sensor probe and/or heating filament, as described above with respect to the inspiratory tube **106**, integrated with it to reduce the opportunity for condensation. Furthermore, the expiratory tube **120** need not return exhaled gases to the gases source **102**. Alternatively, exhaled gases may be passed directly to ambient surroundings or to other ancillary equipment, such as an air scrubber/filter (not shown). In certain embodiments, the expiratory tube **120** is omitted altogether.

(33) With reference to FIG. 2, a humidification apparatus **150** that can be used as the humidifier **104** in the system **100** is shown. As illustrated, a connector **152** pneumatically connects the distal end **110** of the inspiratory tube **106** to the port (not shown) of the humidification chamber **134**. The connector **152** may also facilitate electrical connection to the humidifier **104**. The conduit **106** may comprise one or more resistive heating wires that provide for heating of gases flowing through the conduit **106** and/or sensor wires that electrically or otherwise facilitate communication of signals relating to one or more parameters of the system **100**. Thus, the term “electrical connection” is used to distinguish from “pneumatic connection” and should not be construed in a limiting way.

(34) The humidifier **104** further includes a panel **154**. The panel **154** may be used to mount a user display and/or controls. For example, various dials, switches, and other input means may be used to control operation of the device. Additionally or alternatively, a touch screen display may be used. The user display may display parameters of the system **100**, warnings in the event of any errors or malfunctions, or prompts where user action is required, etc. Where a touch screen display is used, the same display may be used to present information to a user and receive inputs from a user, at least in part.

(35) As illustrated in FIG. 2, the conduit **106** is mounted to the humidification chamber **134** such that the conduit **106** extends substantially parallel to the direction of motion of the humidification chamber **134** as it is slid on or off of the humidifier **104**. With reference to FIG. 3, a sectioned view of a portion of the humidifier **104** is shown. As shown, a probe **156** can be positioned within the outlet port **112** upstream of the connector **152** that connects the chamber **134** to the inspiratory tube **106**. In some configurations, the probe **156** can be positioned in the port **112**. In some configurations, the probe **156** can be positioned within the connector **152**. In some configurations, the port **112** is formed as an elbow with a vertically extending portion and a horizontally extending portion. Other configurations are possible.

(36) In configurations with the elbow port **112** having a sharp corner, it has been discovered that a significant amount of recirculation occurs in the horizontal portion of the port **112**. More particularly, with reference to FIGS. 7-12, a sharp corner at the inside lower transition from vertical to horizontal causes a significant amount of recirculation within the flow. This recirculation, in effect, creates a significant region of dead space. The gases flow can cool within the recirculation/dead space region, which can cause condensation to occur within this region.

(37) FIGS. 7-12 show examples of the velocity and temperature profiles that can be generated for flow rates of 5 L/min, 30 L/min and 60 L/min respectively. The system **100** is not limited to these flow rates; they merely provide an indication of the relationship between different flow rates and the generated fluid characteristics. It appears that a larger dead space region may be generated at higher flow rates.

(38) The system **100** disclosed herein may improve the fluid characteristics of the respiratory gas and may reduce the amount of condensate or other fluid that flows back to the humidification chamber **134** from the attached conduit **106**. To improve the fluid characteristics in the system **100**, the angle of the elbow port **112** of the humidification chamber **134** may be increased; this may soften the sharp corner that is shown in FIGS. 7-12.

(39) With reference again to FIG. 3, a raised portion (or insert) **160** can be attached to, or integrally formed with, one or more components of the system **100**. For example, the raised portion **160** is

shown in FIG. 3 as being positioned within the connector **152**. By way of another example, the raised portion **160** is shown in FIG. 4 as being positioned within the port **112**. In some configurations, the raised portion **160** can be positioned within the inspiratory conduit **106** or the like. The raised portion **160** may be inserted or added to a component using moulding techniques, adhesives, or other appropriate ways of attachment.

(40) In some configurations, an elbow portion **162** can be defined at the transition from a first direction to a second direction of the port **112**. In some configurations, the elbow portion **162** is defined by a transition from a first direction to a second direction that is generally normal to the first direction. In some configurations, the elbow portion **162** is defined by a transition from vertical to horizontal.

(41) As shown in FIG. 3 and FIG. 4, the raised portion **160** can be positioned directly adjacent to the elbow portion **162**. As used herein, directly adjacent has its ordinary meaning of sharing a border or boundary. In some configurations, the raised portion **160** is positioned within the gas flow of only one of the two portions of the port **112**. In other words, in the illustrated configurations, the raised portion **160** protrudes into the gas flow path of the horizontal portion but does not protrude into the gas flow path of the vertical portion. In some configurations, at least the upstream end **164** of the raised portion **160** is positioned entirely within a region that would be a recirculation region without the raised portion **160** present. In some such configurations, both the upstream end **164** and the downstream end **166** of the raised portion **160** are positioned within a region that would be a recirculation region without the raised portion **160** present. In some configurations, the entire raised portion **160** is positioned within a region that would be a recirculation region without the raised portion **160** present. In some configurations, the raised portion **160** is located in the gases flow path directly adjacent to a sharp corner **170** created by the angle of the elbow portion **162** disposed along the gases flow path within a region defined between the humidification chamber **134** and the conduit **106**. In some configurations, the sharp corner **170** is positioned along the gases flow path at a location where the gases flow path makes an abrupt change in direction.

(42) With reference now to FIG. 5 and FIG. 6, the raised portion **160** may resemble a speed bump, protrusion, insert, or the like. The raised portion **160** can be integrally formed with the port **112**, the connector **152**, or another component that joins the humidification chamber **134** and the inspiratory conduit **106**. The raised portion **160** can be a separate and/or separable component from the port **112**, the connector **152**, or other component and secured thereto in any suitable manner.

(43) As shown in FIG. 4, the raised portion **160** is attached to the outlet port **112** of the humidification chamber **134**. The outlet port **112** comprises the elbow portion **162**, which, in this case, creates an angle of, but not limited to, 90°. The raised portion **160** is located at or near the sharp corner **170** created by the angle of the elbow portion **162** because the fluid may be more likely to follow the surface of the port **112** if it is a smooth curve.

(44) The size of the raised portion **160** can be determined by the temperature and velocity profiles, examples of which are shown in FIGS. 7-12, which estimate the size of a dead space region **172** present in the system **100** at high flows. As discussed above, the dead space region **172** may appear to be larger at higher flow rates and, therefore, if the raised portion **160** is made to accommodate these flow rates, it may also accommodate lower flow rates. Occlusion of flow is less likely to occur at lower flow rates and, because the raised portion **160** fills the regions of otherwise dead space present in the system **100**, there are minimal limitations of flow. Thus, the raised portion **160** may accommodate higher flow rates, although, in some embodiments, lower flow rates may also be accommodated. The system **100** discloses the use of the raised portion **160** to accommodate flows between about 5 L/min and about 60 L/min. Some embodiments may accommodate flow rates that exceed these values, for example between 0 L/min and about 100 L/min.

(45) FIG. 3 shows the raised portion **160** inserted into the port **112** as the connector **152** is connected with the humidification chamber port **112**. The raised portion **160** may be attached to the connector **152**, the raised portion **160** may be attached to the conduit **106** or, in some embodiments,

the raised portion **160** may act as a tongue that may be attached to the connector **152** or the conduit **106** and that may extend past the edge of the connector **152** into the outlet port **112**. In some embodiments, the connector **152** may be a male connector that is used with the raised portion **160** attached at the edge of the connector **152**. This is not considered to be an exhaustive list of mechanisms to insert the raised portion **160** into the port **112** of the humidification chamber **134**, but merely a listing of examples of mechanisms that could be used, recognising that other mechanisms may also be possible.

(46) The raised portion **160** may be located near the sharp corner **170** to soften the sharp corner **170** and to fill at least a portion of, if not the entirety of, the dead space region **172** that is formed (see FIGS. 7-12). Benefits of inserting the raised portion **160** into the port **112** of the humidification chamber **134** as the connector **152** is inserted include the reduced distance between the outlet port **112** of the humidification chamber **134** and the conduit **106**. This may be beneficial because the connector **152** itself is unheated, which allows the gas to cool therein, which can cause the formation of condensate as the gas travels between the outlet port **112** of the humidification chamber **134** and the conduit **106**. By reducing the distance between the outlet port **112** of the humidification chamber **134** and the heated portion of the conduit **106**, the time available for gas cooling may be reduced. As a result, the amount of condensate that is formed may also be reduced.

(47) The raised portion **160** may also act to occlude some or all of any condensate or other liquid flowing back into the humidification chamber **134**. By filling at least a portion of the dead space **172**, the lower temperature zone of recirculation can be reduced and, thus, it is likely that less condensate is generated due to the reduced flow separation that may be produced. This may improve the temperature profile of the gas over the unheated connector **152** as it moves towards the heated conduit **106**, which may result in less temperature loss as it passes through the unheated connector **152**.

(48) FIG. 5 and FIG. 6 show possible cross sections of the raised portion **160**, such as that discussed above. FIG. 5 shows the raised portion **160** having a tapered edge while FIG. 6 shows the raised portion **160** having a straight edge. Other configurations also are possible.

(49) With reference to FIGS. 13(a)-13(d), several different configurations that have been investigated are illustrated. Note that FIGS. 13(a)-13(d) are depictions of the gases flow path and, as such, the raised portion **160** is not explicitly shown and the conduit **106**, the connector **152**, and the port **112** also are not explicitly shown. Rather, these illustrations simply reflect the gas flow path defined by these boundary components.

(50) FIG. 13(a) illustrates a configuration without any raised portion. FIG. 13(a), however, is useful to establish a baseline and to explain some of the dimensions used in the following discussion. As illustrated, the flow path has an outlet diameter d_1 and an inlet diameter d_2 . In some configurations, the outlet diameter and the inlet diameter are the same. In some configurations, the outlet diameter d_1 is larger than the inlet diameter d_2 (i.e., the conduit **106** has a larger bore diameter than the port **112**). In the illustrated configurations, the inlet diameter d_2 is 20 mm while the outlet diameter d_1 is 21 mm. Also, in the illustrated configurations, the radius r between the inlet portion and the outlet portion is 10 mm. In each of the configurations shown in FIGS. 13(a)-13(d), these basic dimensions are the same.

(51) FIG. 13(b) illustrates a smaller asymmetric embodiment of the raised portion **160**. FIG. 13(c) illustrates a larger asymmetric embodiment of the raised portion **160**. FIG. 13(d) illustrates a symmetric embodiment of the raised portion **160**. With reference to FIGS. 13(b), 13(c), and 13(d), each of the raised portions **160** can be generally defined in section with the illustrated dimensions. Each of the raised portions **160** will have a first angle (α) at the downstream end **166**, a second angle (β) at the upstream end **164**, a length (w) and a height (h). A volume reduction in the flow path in the region of the raised portions **160** can be defined as the volume of the raised portion **160** divided by the volume of the cylinder bounded by the length (w) of the raised portion **160** and the outlet diameter (d_1).

(52) In some configurations, the first angle (α) can be determined once the other dimensions have been determined. For example, in some configurations, the height (h) can be between 15% and 30% of the outlet diameter d_1 . The second angle (β) can be between 30° and 60°. The length (w) can be larger than 18 mm. Thus, the upper limit of the length (w) and the corresponding range for the first angle (α) can be a function of the other values. In some configurations, however, the first angle (α) can be between 50° and 70°. In some configurations, the first angle (α) can exceed 60° such that condensate is more likely to be contained within the conduit **106** rather than flowing back into the humidification chamber **134**.

(53) With reference now to FIG. **13(b)**, the height (h) of the illustrated raised portion **160** is 4 mm. The length (w) of the raised portion **160** is 25 mm. The first angle (α) of the raised portion **160** is 15 degrees and the second angle (β) is 35 degrees. The illustrated raised portion **160** creates a volume reduction of 4%.

(54) With reference now to FIG. **13(c)**, the height (h) of the illustrated raised portion **160** is 6 mm. The length (w) of the raised portion **160** is 35 mm. The first angle (α) of the raised portion **160** is 15 degrees and the second angle (β) is 60 degrees. The illustrated raised portion **160** creates a volume reduction of 14%.

(55) With reference now to FIG. **13(d)**, the height (h) of the illustrated raised portion **160** is 6 mm. The length (w) of the raised portion **160** is 18 mm. The first angle (α) of the raised portion **160** is 60 degrees and the second angle (β) is 60 degrees. The illustrated raised portion **160** creates a volume reduction of 8%.

(56) FIGS. **14(a)-14(d)** correspond to FIGS. **13(a)-13(d)** and demonstrate the effect on the flow velocity in each configuration. FIGS. **14(a)-14(d)** are flow velocity contours at a flow rate of 5 LPM. The contours of the no raised portion configuration of FIG. **14(a)** clearly show that the elbow portion **162** creates an area of lower velocity gases, and it is in this area that the gases can cool. Each of the raised portions **160** of FIGS. **14(b)-14(d)** has a positive effect on flow velocity, especially on velocities in the middle of the range. In addition, it can be seen that the more smoothly tapering asymmetric raised portions **160** of FIGS. **14(b)** and **14(c)** have significantly improved flow characteristics, because the shallow angle (α) of the tapering end of the raised portion **160** reduces the area of potential dead space downstream of the raised portion **160**.

(57) FIGS. **15(a)-15(d)** correspond to FIGS. **13(a)-13(d)** and show reverse flow vectors extending from the planes of several cross sections downstream of the elbow portion **162**. These vectors are overlaid onto the flow velocity contours of FIGS. **14(a)-14(d)**. The vector lengths represent the magnitudes of negative flow velocity components (i.e., the velocities of flow in the reverse direction to normal flow), which are indicative of recirculation that results in cooling gases. From these illustrations, it is apparent that the regions having the reverse flow components also have the lowest velocities, so it stands to reason that reducing recirculation will improve overall flow velocity and decrease cooling of the gases. Each of the raised portions **160** reduces recirculation relative to the configuration with no raised portion. The large asymmetric raised portion **160** of FIG. **15(c)** appears to reduce recirculation the most. Again, the symmetric raised portion **160** of FIG. **15(d)** does not appear to improve flow characteristics as much as the asymmetric raised portions **160** of FIGS. **15(b)** and **15(c)**.

(58) FIGS. **16(a)-16(c)** correspond to FIGS. **13(a)-13(c)**; there is no corresponding illustration for FIG. **13(d)**. FIGS. **16(a)-16(c)** show streamlines coloured by velocity magnitude and indicating paths taken by a zero mass particle if that particle were traveling with the flow. As shown in these illustrations, the recirculation and low velocity zones are greater in size following the sharp corner **170** of the elbow portion **162** where there is no raised portion. In addition, the larger raised portion **160** of FIG. **16(c)** appears to improve performance over the smaller raised portion **160** of FIG. **16(b)**.

(59) FIG. **17** is a plot illustrating the flow velocity components in a 5 LPM flow relative to the direction of flow at the same cross section for each of the configurations of FIGS. **13(a)-13(d)**.

Negative velocities indicate flow velocity components in the reverse direction compared to normal flow (i.e., recirculation). Each of the configurations demonstrates very similar velocities in the upper region of the cross-section (i.e., the right side of the plot) but there are significant differences in the lower region of the cross-section (i.e., the left side of the plot). The configuration of FIG. **13(a)**, which has no raised portion, demonstrates large regions of stagnation (near 0) and recirculation (below 0) before reaching the upper region and middle region, where positive flow is taking place. As shown, the larger raised portion **160** of FIG. **13(c)** demonstrates the most improvement to reducing or eliminating recirculation and/or stagnant flow.

(60) FIG. **18** is a plot comparing pressure magnitudes taken along a horizontal line that extends along an axis of the configurations of FIGS. **13(a)-13(d)**. As shown in the plot of FIG. **18**, the pressure drop, and therefore the resistance to flow, remains very similar for all four of the designs of FIGS. **13(a)-13(d)**. As such, the benefits of recirculation reduction can be obtained without significantly impacting resistance to flow.

(61) In addition to improving flow characteristics, the raised portions **160** provide the advantage of thwarting the flow of condensate back into the chamber **134**. In configurations desiring to take advantage of this characteristic, the symmetrical raised portion **160** of FIG. **13(d)** provides enhanced performance over the asymmetrical raised portions **160** of FIGS. **13(b)** and **13(c)**. In addition, configurations as shown in FIGS. **19(a)-19(d)** use different geometries to allow gravity to reduce the likelihood of condensate flowing back to the humidification chamber **134**. In each of the embodiments in FIGS. **19(a)-19(c)**, either the conduit **106** or the connector **152** has a lower region before attaching to the port **112** of the humidification chamber **134**. These regions may not inhibit gas flow, but these regions may allow pooling of any condensate formed while reducing the likelihood of the condensate flowing back to the humidification chamber **134**. FIG. **19(d)** uses an inverted form of the previous embodiments, where a high region is located proximally with a lower more distal region following, to reduce the likelihood of any condensate from flowing back to the humidification chamber **134**.

(62) These embodiments may provide solutions for condensate formation and pooling that may occur as a result of using a horizontal connection mechanism between a humidification chamber and a conduit. It is recognised that other mechanisms to reduce the likelihood of condensate flowing back to the humidification chamber may be possible and are not excluded from the scope of the disclosed apparatus and systems.

(63) Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise”, “comprising”, and the like, are to be construed in an inclusive sense as opposed to an exclusive or exhaustive sense, that is to say, in the sense of “including, but not limited to”.

(64) Reference to any prior art in this specification is not, and should not be taken as, an acknowledgement or any form of suggestion that that prior art forms part of the common general knowledge in the field of endeavour in any country in the world.

(65) The disclosed apparatus and systems may also be said broadly to consist in the parts, elements and features referred to or indicated in the specification of the application, individually or collectively, in any or all combinations of two or more of said parts, elements or features.

(66) Where, in the foregoing description reference has been made to integers or components having known equivalents thereof, those integers are herein incorporated as if individually set forth.

(67) Where a value is used with a term of approximation, that number is intended to include the range of roundable values unless otherwise apparent from the context of use.

(68) It should be noted that various changes and modifications to the presently preferred embodiments described herein will be apparent to those skilled in the art. Such changes and modifications may be made without departing from the spirit and scope of the disclosed apparatus and systems and without diminishing its attendant advantages. For instance, various components may be repositioned as desired. It is therefore intended that such changes and modifications be included within the scope of the disclosed apparatus and systems. Moreover, not all of the features,

aspects and advantages are necessarily required to practice the disclosed apparatus and systems. Accordingly, the scope of the present disclosure is intended to be defined only by the claims that follow.

Claims

1. A humidification chamber for a respiratory humidification system comprising: an inlet port defining at least a first portion of a gases flow path; an outlet port defining at least a second portion of the gases flow path; and a body positioned in the outlet port, the body reducing an internal volume of the outlet port, a maximum height of the body is between 15% and 30% of a downstream diameter of the outlet port.
2. The humidification chamber of claim 1, wherein the body is asymmetrical along a length of the body, the length of the body extending in a direction of the gases flow path.
3. The humidification chamber of claim 1, wherein a maximum length of the body is greater than a maximum width of the body, the maximum length of the body extending in a direction of the gases flow path.
4. The humidification chamber of claim 2, wherein the body comprises a raised section extending smoothly towards a tapered downstream edge.
5. The humidification chamber of claim 1, wherein the outlet port comprises a first section extending in a first direction and a second section extending in a second direction, wherein the second section is downstream of the first section.
6. The humidification chamber of claim 5, wherein an interface between the first section and the second section comprises an abrupt change in direction of the gases flow path.
7. The humidification chamber of claim 5, wherein the first direction is approximately perpendicular to the second direction, such that the outlet port comprises an elbow portion.
8. The humidification chamber of claim 5, wherein the first section comprises an opening configured to receive a sensor.
9. The humidification chamber of claim 6, wherein an upstream edge of the body is positioned near the interface, the body forming a smooth transition between the first section to the second section for the gases flow path in use.
10. The humidification chamber of claim 1, wherein the body fills or at least partially fills a dead space region in the gases flow path in use.
11. The humidification chamber of claim 1, wherein the body reduces regions of low velocity along the gases flow path.
12. The humidification chamber of claim 1, wherein the body reduces recirculation of gases within the outlet port.
13. The humidification chamber of claim 1, wherein the body reduces stagnant flow within the gases flow path.
14. The humidification chamber of claim 1, wherein the body reduces an amount of condensate generated within the gases flow path in use, wherein the body reduces liquid flowing from a conduit connected to the outlet port to the humidification chamber in use.
15. The humidification chamber of claim 1, wherein the body reduces regions of low temperature along the gases flow path.
16. The humidification chamber of claim 1, wherein an angle between an upstream edge of the body and the outlet port is between 30 degrees and 60 degrees.
17. The humidification chamber of claim 1, wherein the body comprises an insert configured to be removably attached to the outlet port.
18. The humidification chamber of claim 1, wherein the body provides a volume reduction in the outlet port of approximately 14%.

19. The humidification chamber of claim 1, wherein the body is integrally formed with the outlet port.
